Higher National Diploma in Information Technology

First Year First Semester Examination – 2023

HNDIT 1032-Computer & Network System(New)

Q1)

(i)Briefly **explain the following** terms.

a. **Analog Signal**.

b. **Digital Signal**.

(iii)Summarize the main technologies of following computer generations. **(06 marks)**

a. 1st Generation.

b. 2nd **Generation**.

c. **4th Generation**.

(iii)What are the **key** characteristics **of followings**.(04 **marks)**

a. **Microcomputer**.

b. **Mainframe**.

(iv)Explain the usages of computers **in following areas**.

(06 marks)

a. Education.

b. Banking.

c. Government

Q2)

(i)Define following data units.

a. Nibble.

b. **Word.**

(Total 20 marks)

(ii)What is non positional **number** system**?**

(02 marks)

(iii)Perform following calculations.

(12 marks)

a. **13510 to Binary**.

b. 42510 **to Octal**.

c. 75410 **to Hexadecimal**.

d. 101111012 **to Decimal**.

e. 6258 to Binary.

f. ABC16 to Octal. .

(iv)Identify the key characteristics following concepts.

(04 marks)

a. **ASCII**.

b. **Unicode**.

Q3)

(i)Briefly explain the following circuits **with examples**.

a. **Sequential circuit**.

b. **Combinational circuit**.

(ii)"Half Adder is one of **the** fundamental **circuits** inside **the CPU**." **(04 marks)**

a. **Draw the circuit diagram**.

b. **Draw the** truth **table.**

(iii)Write down **the Boolean expression of following laws**. **(08 marks)**

a. **Association**

b. Commutative.

c. Distributive.

d. **De Morgan**

(iv)Draw the Karnaugh map and **simplify the circuit**.

Q4)

(i)Define **the following CPU components?**

a. **ALU**

b. **Register**

(ii)Briefly explain the following concepts.

a. **Fetch**

b. **Decode**

c. Execute

(iii)Give examples on following memory types.

a. **DRAM**.

b. SRAM.

c. **Volatile** memory.

d. **Nonvolatile memory.**

(iv)Calculate the hard disk capacity with following information.

512 **bytes/sector**
300 sectors/track (on average)
20,000 **tracks/surface**
2 surfaces/platter
5 platters/disk

(Total 20 marks)

Q5)

(i)Write down the functions of **Operating System**.

(04 marks)

(ii)Define the following terms.(04 **marks)**

a. Multitasking OS.

b. Embedded OS.

(iii)The **motherboard** is **the** hub, **which is used** to connect **all the** essential components **of** a computer. **Briefly explain the interfaces and** port **with examples**.

(04 marks)

(iv)Briefly describe System Buses .

(v)Briefly explain following system **utilities**.

a. **Disk** Defragment

b. **Event viewer**

Q6)

(i)Write down the advantages **of Computer Network?**

(ii)Define following terms.

a. **LAN**

b. **WAN**

(iii)Identify the functions **of following** network devices.

a. **NIC.**

b. **Repeater.**

c. Switch**.**

d. Router.

(iii)Draw the diagrams to represent following topologies.

a. **Bus.**

b. Star.

(v)What are main functionalities **of following** OSI layers**.**

a. **Physical.**

b. **Network.**

c. Transport.

d. **Application.**